

1. Question: What are arithmetic operators in Python, and can you provide examples?

Answer:

Arithmetic operators in Python are used for performing mathematical operations. The basic arithmetic operators are:

- + (Addition): Adds two operands. Example: `a + b`
- (Subtraction): Subtracts the right operand from the left operand. Example: `a - b`
- * (Multiplication): Multiplies two operands. Example: `a * b`
- / (Division): Divides the left operand by the right operand. Example: `a / b`
- % (Modulus): Returns the remainder of the division. Example: `a % b`
- // (Floor Division): Returns the quotient, discarding the remainder. Example: `a // b`
- ** (Exponentiation): Raises the left operand to the power of the right operand. Example: `a ** b`

2. Question: How do assignment operators work in Python?

Answer:

Assignment operators are used to assign values to variables. The basic assignment operator is `=`. There are also compound assignment operators that combine an operation with assignment:

- `+=`: Adds the right operand to the left operand and assigns the result to the left operand. Example: `x += 3` is equivalent to `x = x + 3`.
 - `-=`: Subtracts the right operand from the left operand and assigns the result to the left operand. Example: `x -= 2` is equivalent to `x = x - 2`.
- Other examples include `*=`, `/=`, etc.

3. Question: What are comparison operators in Python and what do they return?

Answer:

Comparison operators are used to compare two values. They return a Boolean value (True or False). The basic comparison operators are:

- `==` (Equal to): Returns True if the operands are equal. Example: `a == b`
- `!=` (Not equal to): Returns True if the operands are not equal. Example: `a != b`
- `<` (Less than): Returns True if the left operand is less than the right operand. Example: `a < b`
- `>` (Greater than): Returns True if the left operand is greater than the right operand. Example: `a > b`
- `<=` (Less than or equal to): Returns True if the left operand is less than or equal to the right operand. Example: `a <= b`
- `>=` (Greater than or equal to): Returns True if the left operand is greater than or equal to the right operand. Example: `a >= b`

4. Question: Can you explain the logical operators in Python with examples?

Answer:

Logical operators in Python are used to combine conditional statements. The basic logical operators are:

- `and` (Logical AND): Returns True if both conditions are true. Example: `x and y`
- `or` (Logical OR): Returns True if at least one condition is true. Example: `x or y`
- `not` (Logical NOT): Inverts the Boolean value of the operand. Example: `not x`

1. What is base and exponent in exponentiation?

- In exponentiation, the base refers to the number that is raised to a certain power, while the exponent (or power) refers to the number of times the base is multiplied by itself. For example, in the expression 2^3 , 2 is the base and 3 is the exponent. It means 2 is multiplied by itself 3 times ($2 \times 2 \times 2 = 8$).

2. What is a compound assignment operator?

- A compound assignment operator is a shorthand notation that combines an arithmetic or bitwise operation with assignment. For instance, instead of writing $x = x + 5$, you can use the compound assignment operator $+=$, like $x += 5$, which means add 5 to the current value of x and store the result back in x. Similarly, you have $-=$ for subtraction, $*=$ for multiplication, $/=$ for division, and so on.

3. Difference between division and floor division.

- In Python, the $/$ operator performs regular division, yielding a floating-point result. For example, $5 / 2$ would result in 2.5.

- The $//$ operator performs floor division, which always rounds down to the nearest whole number. For example, $5 // 2$ would result in 2. It truncates any fractional part and returns the integer part of the division result.

- So, the key difference is that regular division ($/$) returns a floating-point result, while floor division ($//$) returns an integer result, truncating any fractional part.